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Stress hyperglycemia ratio modulates the impact of organ dysfunction on mortality risk in critically ill patients with atrial fibrillation: insights from the MIMIC-IV database

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Abstract

Background Atrial fibrillation (AF) is a major cardiovascular issue in critically ill patients, linked to elevated mortality rates. The Stress Hyperglycemia Ratio (SHR), a novel metric of glucose control, has shown promise in predicting adverse outcomes in cardiovascular diseases. However, its impact on the relationship between organ dysfunction and mortality in AF remains unclear. This study aims to explore SHR's role in modifying this association to improve risk prediction in critically ill AF patients.

Methods A retrospective cohort study was performed on MIMIC-IV 3.1 patients with critical illness and AF. The primary endpoints were in-hospital, 30-day, and 1-year mortality. Patients were stratified by SHR quartiles and categorized into four groups based on combined SHR and SOFA scores. Kaplan-Meier survival analysis and multivariate Cox regression assessed the association with mortality, and restriction cubic splines (RCS) were used to examine non-linear relationships. Mediation analysis using the Bootstrap method quantified the role of serum bicarbonate. A nomogram was developed with multivariable logistic regression, incorporating SHR, SOFA, and significant covariates from backward selection. The SHapley Additive exPlanations(SHAP) analysis provided insights into predictor contributions.

Results Of the 15,358 patients included in the cohort, the median age was 66 years (IQR: 55–76), with 56.8% being male. The SHR-SOFA combination significantly stratified patients into four distinct risk groups (Q1-Q4), with the high SHR and high SOFA group (Q4) showing the poorest prognosis at all time points (log-rank $P < 0.001$). Multivariate Cox regression confirmed that Q4 had the highest mortality risk across all time points (in-hospital HR: 5.32, 95% CI: 4.31–6.57; 30-day HR: 4.27, 95% CI: 3.62–5.04; 1-year HR: 3.18, 95% CI: 2.82–3.59). RCS analysis indicated a nonlinear correlation between SHR and mortality, with a threshold effect at SHR = 1.0 in low SOFA patients. In contrast, high SOFA patients showed a J-shaped curve, with an ideal SHR of 1.2. The predictive nomogram, validated through backward stepwise regression, included SHR, SOFA score, age, mean arterial pressure (MBP), white blood cell count (WBC), bicarbonate, hemoglobin, sodium, and total calcium as key predictors. SHAP analysis highlighted SHR and

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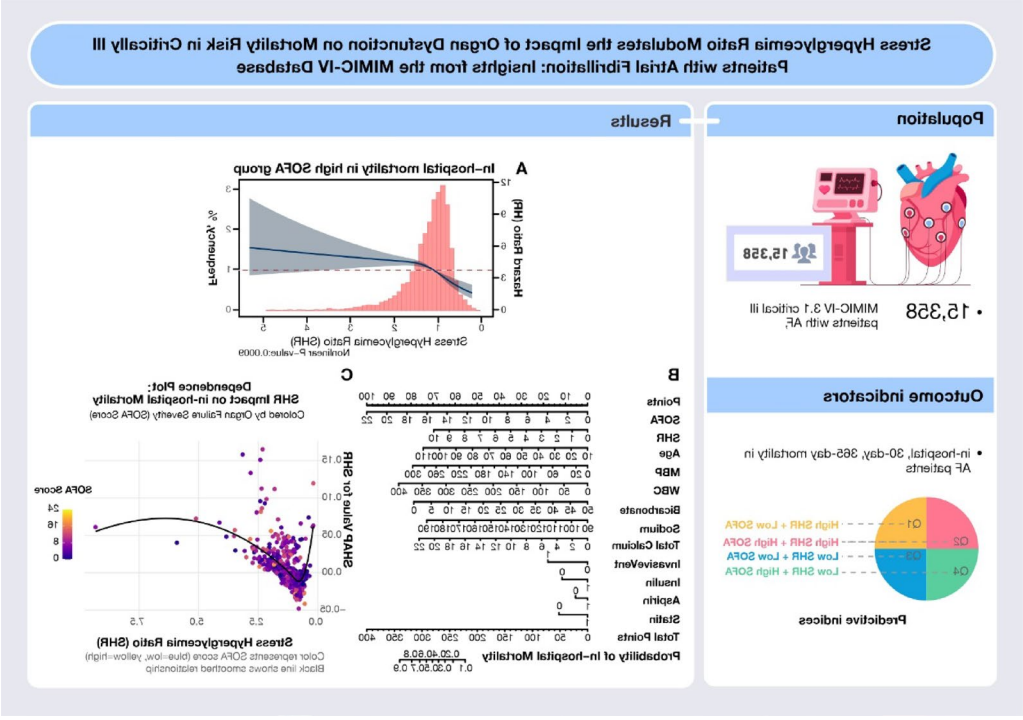
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SOFA as the most influential predictors, with age also playing a crucial role. Mediation analysis indicated that serum bicarbonate mediated 22.7% of SHR's effect on in-hospital mortality, diminishing over time.

Conclusions This study reveals a notable relationship among SHR and SOFA scores in forecasting mortality among critically sick patients with atrial fibrillation. The developed nomogram integrating SHR, SOFA, and other covariates offers a novel tool for risk stratification. SHAP analysis confirms SHR and SOFA as key predictors, highlighting their clinical utility in mortality assessment.

Keywords Atrial fibrillation, Stress hyperglycemia ratio, Sequential organ failure assessment, Mortality prediction, Machine learning interpretation

Graphical abstract



Research insights

What is currently known about this topic?

- SHR predicts cardiovascular events in critical illness.
- SOFA scores predict mortality but ignore metabolic dysregulation.
- AF mortality links to organ failure yet metabolic modulation unquantified.

What is the key research question?

- How does SHR modulate organ dysfunction's impact on mortality in critical AF?

What is new?

- This study delineates a nonlinear correlation between SHR and mortality across various SOFA levels.
- We provide a predictive nomogram integrating SHR and SOFA for mortality risk assessment.
- Insights into the mediating role of bicarbonate in the SHR-mortality link.

How might this study influence clinical practice?

- The nomogram can help clinicians stratify mortality risk in critically ill AF patients;
- Understanding the SHR-SOFA interaction may guide targeted interventions.

Introduction

Atrial fibrillation (AF) ranks as one of the most prevalent persistent arrhythmias associated with cardiovascular disease [1]. The prevalence of AF is estimated to be around 330 million people worldwide, with nearly one third of these individuals developing this disorder during their lifetime [2]. AF is associated with a 1.5 to 3.5-fold

increase in all-cause mortality, representing a substantial medical and economic burden worldwide. Within intensive care units (ICUs), the incidence of AF among critically ill patients has been reported to range from 4.5% to 29.5% [3–5]. It is important to note that AF in the ICU may represent either a primary reason for admission (e.g., arrhythmia-induced hemodynamic instability) or a secondary complication arising from acute illness, sepsis, metabolic disturbance, or postoperative stress. Nevertheless, prognostic studies seldom distinguish between these contexts, and there remains a notable gap in research dedicated to identifying specific risk factors—particularly metabolic markers—associated with outcomes in critically ill AF patients. Therefore, the identification and management of risk factors are crucial for mitigating mortality rates in this vulnerable population.

Current research demonstrates that hyperglycemia is common among critically ill patients and correlates with a higher incidence of complications and mortality, regardless of diabetes status [6–8]. A new metric of glucose control, the Stress Hyperglycemia Ratio (SHR), serves as a comprehensive assessment technique that balances acute and chronic blood glucose levels, therefore providing a more realistic reflection of stress-induced hyperglycemia, has been proposed to play a significant role in hyperglycemia-related processes [9]. In the full range of glycemic conditions, SHR has been proposed as a promising indicator of stress and hyperglycemia. Hyperglycemia induced by acute stress occurs when glucagon, cortisol, catecholamines, and growth hormone levels are elevated. In addition to increasing gluconeogenesis and glycogenolysis, these processes also inhibit peripheral glucose uptake. SHR serves as a standalone prognostic factor for negative cardiovascular events, including heart failure and AF, following acute myocardial infarction (AMI) or after cardiac surgical procedures [10–12]. Effective glycemic management is crucial in the treatment of AF [13], and recent research suggests that SHR correlates with both short-term and long-term mortality rates in patients experiencing severe AF [14]. However, while SHR's U-shaped mortality association is established, a cardinal extension is quantifying its modification of organ dysfunction impact. Concomitantly, the biochemical mediation through acid-based imbalance remains unexplored terrain, culminating in deficient time-sensitive prediction tools.

The purpose of this study was to examine how SHR modifies the association between organ dysfunction severity that is assessed by SOFA scores and mortality risk in critical atrial fibrillation. Additionally, this study quantified bicarbonate-mediated pathways, constructed time-stratified prediction models and utilized explainable machine learning to characterize interaction patterns.

Method

Data source

This retrospective cohort analysis used data from MIMIC-IV (version 3.1), a publicly accessible deidentified clinical database managed by the Massachusetts Institute of Technology Computational Physiology Laboratory [15]. From 2008 to 2022, over 90,000 ICU admissions were documented in the MIMIC-IV database at Beth Israel Deaconess Medical Center in Boston, Massachusetts. Data captured by MIMIC-IV includes demographic information, continuous bedside vital measurements, laboratory test results, medication records, surgical and procedural interventions, clinician notes, and ICD-9 and ICD-10 disease diagnoses, as well as longitudinal survival outcome data. As a result of structural improvements and expanded data components, the analysis employed MIMIC-IV version 3.1. BIDMC and MIT waived informed consent for this retrospective, anonymized database study since it involved retrospective data. Prior to accessing the data, the author HJJ received training in human subject protection and successfully met the Collaborative Institutional Training Initiative's requirements (approval No. 61632129).

Study population

Initially, this retrospective cohort investigation identified 22,419 adult patients (≥ 18 years) who met the following criteria from the MIMIC-IV database and who had documented atrial fibrillation: (1) Admission to an intensive care unit for the first time (2) ICU stay exceeding 24 h. After exclusion criteria for incomplete metabolic data (HbA1c or fasting blood glucose within 24 h of admission) and unavailability of follow-up records, the analysis cohort consisted of 15,358 patients. Using the SHR, these patients were categorized into three groups (Figure S1).

Data extraction

The data is extracted using Navicat Premium (v16.1.15). The correlations between demographic variables (age, gender, weight), vital signs monitoring parameters (heart rate, mean blood pressure, respiratory rate, and oxygen saturation) and validated severity assessment scores (Simplified Acute Physiology Score II and Sequential Organ Failure Assessment) were systematically identified. Comorbidities include hypertension, diabetes, renal disease, coronary heart disease, acute myocardial infarction, and cardiac failure. Lab tests after admission included glucose, HbA1c, lactate, lipid profiles, renal/hepatic function indicators, hematologic parameters, and electrolyte panels. Antibiotics, glucocorticoids, vasopressors, and mechanical ventilation were evaluated. Outcome tracking assessed intensive care unit and hospital stay length and death endpoints from in-hospital to 365 days after discharge. Variables with 20% missing were

eliminated for data completeness. Multiple imputation was performed for variables with < 20% missing values using the Miss Forest method, a non-parametric random forest technique based on the missing-at-random assumption [16, 17]. A directed cyclic graph model was used to identify various causal pathways prior to statistical analysis. The SHR was determined using the formula: fasting blood glucose (FBG) (mg/dl) divided by [28.7 x HbA1c (%) – 46.7].

Study endpoint

The objectives of this study encompassed mortality rates at several intervals after hospitalization: in-hospital, 30 days post-admission and 365 days post-admission. Data regarding patient mortality post-hospital discharge was sourced from the United States Social Security Administration's Death Master File.

Statistical analysis

The SHR was stratified into tertiles for analysis: Tertile 1 (Low SHR: <0.93), Tertile 2 (Medium SHR: 0.93–1.22), and Tertile 3 (High SHR: ≥1.22). The SOFA score was dichotomized using a cutoff of 5, where scores < 5 defined Low SOFA and scores ≥ 5 defined High SOFA, a threshold validated to signify significant organ dysfunction [18]. For the combined analysis, patients were categorized into four groups: Q1 (Low SHR + Low SOFA), Q2 (High SHR + Low SOFA), Q3 (Low SHR + High SOFA), and Q4 (High SHR + High SOFA).

The analytical approach utilized a multi-tiered framework, commencing with thorough data preprocessing. Continuous variables were expressed as mean ± standard deviation for data with normal distribution, while non-normally distributed variables were represented as median with an interquartile range (IQR). Distribution patterns were assessed using the Shapiro-Wilk test. Between-group comparisons were performed using Student's T-tests or Kruskal-Wallis tests as suitable. Categorical variables were summarized using frequencies (%) and analyzed through χ^2 or Fisher's exact tests. Data with missing values exceeding a 20% threshold were excluded, while the remaining missing values were addressed through multiple imputation utilizing the miss Forest algorithm, based on the missing-at-random assumption.

The primary inferential analyses employed Cox proportional hazards regression, advancing all candidate predictors with univariate associations ($P < 0.05$) to multivariate modeling, and the results were presented as an hazard ratio (HR) with 95% confidence intervals (CIs). Hierarchically adjusted models were constructed. Model 1 was based solely on SHR; Model 2 applied extensive adjustments incorporating laboratory markers, therapeutic interventions, and further severity metrics. The assumptions of proportional hazards were validated using Schoenfeld residual analysis. Nonlinear exposure-outcome relationships were assessed using restricted cubic splines (RCS) with four knots. Significant threshold effects ($P < 0.05$ by likelihood ratio test) were quantified through piecewise Cox regression. Survival curves were constructed using the Kaplan-Meier method, with stratification applied according to SHR tertiles and SOFA scores categorized at the median. Survival distribution differences were assessed using log-rank tests across three mortality endpoints: in-hospital, 30-day, and 1-year mortality.

With the aid of the counterfactual framework, a bootstrap-based mediation analysis (1,000 random samples) was conducted to quantify the mediating role played by bicarbonate in the SHR-mortality pathway. The effects were decomposed as direct (c), indirect (a + b), and proportion-mediated effects. In the process of developing and validating the prognostic tool, we created a nomogram that integrates key predictors identified in the final multivariate models. This enables tailored risk assessment. Bootstrapping-corrected calibration curves were used to verify calibration (Brier scores with bootstrap validation and 200 resamples) to ensure alignment between predictions and observations [19]. Additionally, to enhance the clinical utility of the nomogram developed for predicting mortality risk in critically ill atrial fibrillation patients, we conducted a comprehensive evaluation integrating SHAP (Shapley Additive Explanations) analysis [20]. The use of this method permitted a detailed understanding of how various features affected predictions, increasing clarity and comprehensibility. All analyses were conducted using R v4.3.2 (survival, rms, glmnet packages) with statistical significance established at two-sided $P < 0.05$.

Results

Baseline characteristics

The study assessed 15,358 critically ill patients with atrial fibrillation according to the established inclusion and exclusion criteria (Figure S1), with a median (IQR) age of 66 [55,76] years, and 56.8% being males. The study patients were categorized into three groups based on the SHR index at ICU admission: T1 (< 0.93), T2 (0.93 ≤ SHR < 1.22), and T3 (SHR ≥ 1.22). Table 1 delineates the baseline features of these groups. The elevated SHR group exhibited increased respiratory rates, severity ratings (e.g., SOFA and SIRS), more acid-base imbalances (e.g., reduced bicarbonate), and a higher prevalence of comorbidities (e.g., myocardial infarction, congestive heart failure, and chronic kidney disease). Furthermore, all-cause mortality following ICU admission in-hospital, at 90 days, and at 360 days displayed a gradual rise across SHR tertiles (T1 to T3, $P < 0.001$).

Table 1 Patients' characteristics

Variable	Overall	Tertile 1	Tertile 2	Tertile 3	*P* value
	(*n* = 15358)	(SHR < 0.93, *n* = 5122)	(0.93 ≤ SHR < 1.22, *n* = 5119)	(SHR ≥ 1.22, *n* = 5117)	
Demographics					
Age, years	66.0 [55.0, 76.0]	66.0 [55.0, 77.0]	66.0 [56.0, 76.0]	66.0 [55.0, 76.0]	0.040
Men, n (%)	8720.0 (56.8)	2908.0 (56.8)	2979.0 (58.2)	2833.0 (55.4)	0.015
Race/ethnicity, n (%)					< 0.001
Asian	545.0 (3.5)	186.0 (3.6)	188.0 (3.7)	171.0 (3.3)	
White	9323.0 (60.7)	3076.0 (60.1)	3206.0 (62.6)	3041.0 (59.4)	
Black	2138.0 (13.9)	804.0 (15.7)	641.0 (12.5)	693.0 (13.5)	
Others	3352.0 (21.8)	1056.0 (20.6)	1084.0 (21.2)	1212.0 (23.7)	
Vital signs					
Weight, kg	81.0 [68.2, 97.0]	80.2 [67.3, 95.6]	81.00 [69.0, 96.8]	81.8 [68.8, 98.4]	< 0.001
Temperature, ℃	36.7 [36.4, 37.1]	36.7 [36.4, 37.0]	36.7 [36.4, 37.1]	36.7 [36.4, 37.1]	0.032
Mean heart rate, beats/min	86.0 [75.0, 99.0]	86.0 [75.0, 98.0]	87.0 [75.0, 99.0]	86.0 [74.0, 99.0]	0.074
Systolic blood pressure, mmHg	127.0 [111.0, 145.0]	127.0 [111.0, 144.0]	128.0 [112.0, 145.0]	127.0 [110.0, 145.0]	0.066
Diastolic blood pressure, mmHg	70.0 [59.0, 82.0]	70.0 [59.0, 82.0]	70.0 [59.0, 83.0]	69.0 [58.0, 82.0]	0.117
Respiratory rate, times/min	18.0 [15.0, 22.0]	18.0 [15.0, 22.0]	18.0 [15.0, 22.0]	19.0 [16.0, 23.0]	< 0.001
Saturation of pulse oxygen, %	98.0 (95.0, 100.0)	98.0 [96.0, 100.0]	98.0 [95.0, 100.0]	98.0 [95.0, 100.0]	< 0.001
Comorbidities					
Hypertension, n (%)	6987.0 (45.5)	2379.0 (46.4)	2468.0 (48.2)	2140.0 (41.8)	< 0.001
Diabetes mellitus, n (%)	6401.0 (41.7)	2201.0 (43.0)	1725.0 (33.7)	2475.0 (48.4)	< 0.001
Acute myocardial infarction, n (%)	3255.0 (21.2)	984.0 (19.2)	1047.0 (20.5)	1224.0 (23.9)	< 0.001
Heart failure, n (%)	4501.0 (29.3)	1462.0 (28.5)	1382.0 (27.0)	1657.0 (32.4)	< 0.001
Chronic obstructive pulmonary disease, n (%)	3509.0 (22.8)	1146.0 (22.4)	1140.0 (22.3)	1223.0 (23.9)	0.089
Liver disease, n (%)	1679.0 (10.9)	470.0 (9.2)	500.0 (9.8)	709.0 (13.9)	0.076
Chronic kidney disease, n (%)	3472.0 (22.6)	1160.0 (22.6)	1017.0 (19.9)	1295.0 (25.3)	< 0.001
Stroke, n (%)	4324.0 (28.2)	1504.0 (29.4)	1600.0 (31.3)	1220.0 (23.8)	< 0.001
Cardiogenic shock, n (%)	1151.0 (7.5)	309.0 (6.0)	327.0 (6.4)	515.0 (10.1)	< 0.001
Severity scores					
SOFA	4.0 [2.0, 7.0]	4.0 [2.0, 6.0]	4.0 [2.0, 6.0]	5.0 [3.0, 8.0]	< 0.001
SAPS-II	33.0 [26.0, 41.0]	32.0 [25.0, 40.0]	32.0 [25.0, 40.0]	35.0 [27.0, 44.0]	< 0.001
SIRS	2.0 [2.0, 3.0]	2.0 [2.0, 3.0]	2.0 [2.0, 3.0]	3.0 [2.0, 3.0]	< 0.001
OASIS	30.0 [24.0, 36.0]	29.0 [24.0, 35.0]	30.0 [24.0, 35.0]	31.0 [26.0, 37.0]	< 0.001
Laboratory parameters					
Hemoglobin, g/L	11.6 [9.8, 13.1]	11.6 [10.0, 13.1]	11.7 [10.0, 13.2]	11.3 [9.5, 13.0]	< 0.001
Platelet, 10^9/L	201.0 [150.0, 262.0]	199.0 [150.0, 259.0]	201.0 [153.0, 259.0]	202.0 [149.0, 268.0]	0.880
White blood cell, 10^9/L	10.2 [7.5, 13.9]	9.1 [6.8, 12.3]	10.2 [7.6, 13.5]	11.7 [8.4, 15.8]	< 0.001
Bicarbonate, mmol/L	23.0 [20.0, 25.0]	23.0 [21.0, 26.0]	23.0 [21.0, 25.0]	22.0 [19.0, 25.0]	< 0.001
Creatinine, mg/dL	1.0 [0.8, 1.4]	0.9 [0.7, 1.3]	0.9 [0.7, 1.3]	1.1 [0.8, 1.6]	< 0.001
Blood urea nitrogen, mg/dL	18.0 [13.0, 29.0]	17.0 [12.3, 27.0]	18.0 [13.0, 26.0]	21.0 [14.0, 34.0]	< 0.001
Sodium, mmol/L	138.0 [136.0, 141.0]	139.0 [136.0, 141.0]	138.0 [136.0, 141.0]	137.0 [134.0, 140.0]	0.579
Potassium, mmol/L	4.1 [3.7, 4.5]	4.0 [3.7, 4.4]	4.0 [3.7, 4.4]	4.2 [3.8, 4.6]	< 0.001
Total calcium, mg/dL	8.5 [8.0, 9.0]	8.6 [8.1, 9.0]	8.6 [8.1, 9.0]	8.5 [8.0, 9.0]	< 0.001
Prothrombin time, s	13.5 [12.1, 15.7]	13.5 [12.1, 15.7]	13.4 [12.0, 15.5]	13.5 [12.1, 15.8]	0.013
Activated partial thromboplastin time, s	29.6 [26.1, 36.4]	30.0 [26.4, 36.4]	29.4 [26.2, 35.9]	29.4 [25.7, 36.9]	0.004
Treatment					
Aspirin	6206 (40.4)	2217 (43.3)	2107 (41.2)	1882 (36.8)	< 0.001
Insulin	3723 (24.2)	1345 (26.3)	1005 (19.6)	1373 (26.8)	0.435
Statin	2262 (14.7)	778 (15.2)	753 (14.7)	731 (14.3)	< 0.001
Length Of Stay (LOS)					
LOS in hospital	7.8 [4.7, 13.9]	7.0 [4.3, 12.2]	7.7 [4.7, 13.3]	8.9 [5.2, 16.1]	< 0.001
LOS in ICU	2.6 [1.6, 4.8]	2.2 [1.5, 4.1]	2.5 [1.6, 4.7]	3.0 [1.8, 5.7]	< 0.001
Outcome					

Table 1 (continued)

Variable	Overall	Tertile 1	Tertile 2	Tertile 3	*P* value
In-hospital mortality, n (%)	865 (5.6)	175 (3.4)	239 (4.7)	451 (8.8)	< 0.001
30-day mortality, n (%)	1069 (7.0)	376 (7.3)	489 (9.6)	711 (13.9)	< 0.001
365-day mortality, n (%)	2598 (16.9)	701 (13.7)	805 (15.7)	1092 (21.3)	< 0.001

Table 2 Cox proportional hazard models for in-hospital, 30-day, and 365-day all-cause mortality

Variable	In-hospital mortality			30-day mortality			365-day mortality		
	Model I HR (95%CI)	Model II HR (95%CI)	*P* value	Model I HR (95%CI)	Model II HR (95%CI)	*P* value	Model I HR (95%CI)	Model II HR (95%CI)	*P* value
SHR									
Tertiles 1	Ref	Ref		Ref	Ref		Ref	Ref	
Tertiles 2	1.42 (1.17–1.73)	1.38 (1.13–1.68)	0.0004	1.59 (1.34–1.89)	1.54 (1.30–1.82)	< 0.0001	1.20 (1.09–1.33)	1.20 (1.08–1.33)	0.0004
Tertiles 3	2.84 (2.38–3.38)	1.82 (1.51–2.18)	< 0.0001	2.65 (2.27–3.10)	1.81 (1.54–2.13)	< 0.0001	1.73 (1.57–1.90)	1.38 (1.25–1.53)	< 0.0001
SOFA									
Low	Ref	Ref		Ref	Ref		Ref	Ref	
High	4.17 (3.61–4.81)	2.16 (2.07–2.93)	< 0.0001	3.09 (2.73–3.49)	2.22 (1.91–2.59)	< 0.0001	2.10 (1.94–2.27)	1.88 (1.71–2.08)	< 0.0001
Combined									
Low SHR and Low SOFA	Ref	Ref		Ref	Ref		Ref	Ref	
Low SHR and High SOFA	7.76 (6.40–9.40)	2.59 (2.07–3.23)	< 0.0001	5.12 (4.37-6.00)	2.14 (1.77–2.58)	< 0.0001	2.94 (2.66–3.26)	1.78 (1.58-2.00)	< 0.0001
High SHR and Low SOFA	2.07 (1.63–2.62)	1.79 (1.41–2.28)	< 0.0001	1.66 (1.33–1.97)	1.44 (1.18–1.75)	< 0.0001	1.32 (1.18–1.49)	1.21 (1.08–1.39)	< 0.0001
High SHR and High SOFA	3.92 (3.21–4.79)	4.06 (3.24–5.09)	< 0.0001	2.67 (2.27–3.16)	3.23 (2.75–4.03)	< 0.0001	1.89 (1.70–2.01)	2.45 (2.17–2.78)	< 0.0001

Data: N (%) or Mean (T1–T2) or mean ± standard deviation. Abbreviations: SOFA: Sequential Organ Failure Assessment; SAPS II, Simplified Acute Physiology II; SIRS, Systemic Inflammatory Response Syndrome Score; OASIS, Outcome and Severity Index Score.

The SHR-SOFA combination and all-cause mortality in critical ill patients with AF

After combining the SOFA and the SHR, the patients were divided into four groups: Q1: low SOFA and low SHR; Q2: low SOFA and high SHR; Q3: high SOFA and low SHR; Q4: high SOFA and high SHR. The Kaplan-Meier curves demonstrated statistically significant differences in survival probabilities among the four strata (log-rank $P < 0.001$) (Figure S2) The high SHR and high SOFA group consistently displayed the poorest prognosis at all time points, with intermediate groups maintaining the hierarchy of low SHR and high SOFA > high SHR and low SOFA at all intervals ($P < 0.001$).

Cox regression analysis indicated significantly increased crude mortality hazards at all time points, with the high SHR and high SOFA group displaying the highest risk (in-hospital: HR 5.32, 95% CI 4.31–6.57; 30-day: HR 4.27, 3.62–5.04; 1-year: HR 3.18, 2.82–3.59) compared to the low SHR and low SOFA reference. An association with risk differentials was still observed after adjusting for other variables, compared to the low SHR and low SOFA reference, the low SHR and high SOFA group exhibited HR of 2.59 (95% CI 2.07–3.23) for in-hospital mortality, 2.14 (95% CI 1.77–2.58) for 30-day mortality, and 1.78 (95% CI 1.58-2.00) for 1-year mortality; the high SHR + low SOFA group demonstrated HRs of 1.79 (95% CI 1.41–2.28), 1.44 (95% CI 1.18–1.75), and 1.21 (95% CI 1.08–1.39) for the respective timepoints; while the high SHR and high SOFA group showed substantially elevated hazards (in-hospital: HR 4.06, 95% CI 3.24–5.09; 30-day: HR 3.23, 95% CI 2.75–4.03; 1-year: HR 2.45, 95% CI 2.17–2.78). Analysis of individual components revealed the highest SHR level independently increased in-hospital mortality (HR 1.82, 95% CI 1.51–2.18), 30-day mortality (HR 1.81, 95% CI 1.54–2.13), and 1-year mortality (HR 1.38, 95% CI 1.25–1.53), while high SOFA similarly elevated risks (in-hospital: HR 2.16, 95% CI 2.07–2.93; 30-day: HR 2.22, 95% CI 1.91–2.59; 1-year: HR 1.88, 95% CI 1.71–2.08) (Table 2). Subgroup analysis validated the consistent ranking of mortality risk across cohorts of myocardial infarction, heart failure, and renal failure, reflecting overall population trends with non-significant interaction terms (Table S1-3). For in-hospital mortality, patients aged < 65 years with high SHR and high SOFA had substantially elevated risk (HR 6.93, 95% CI 4.50-10.66) compared to the reference group.

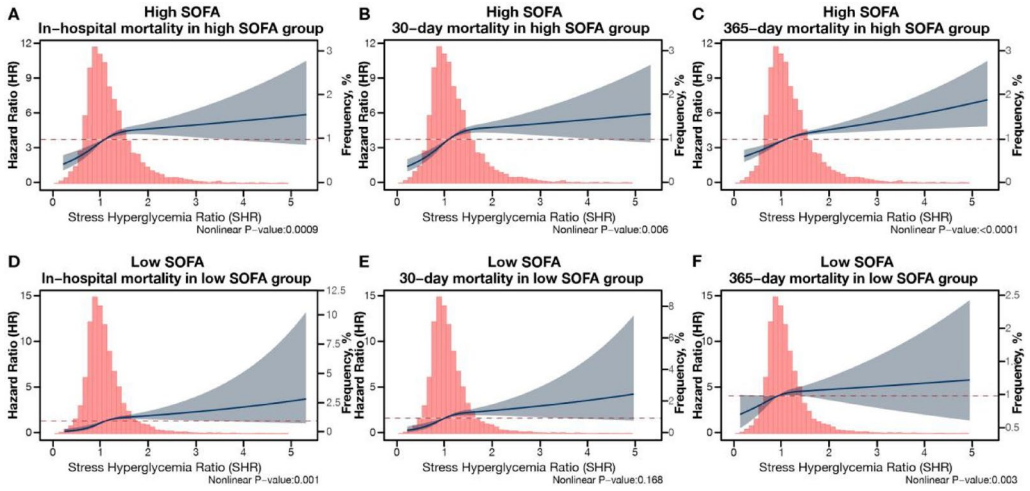


Fig. 1 Restriction Cubic Splines Analysis of Stress Hyperglycemia Ratio on Mortality Risk Across Different SOFA Levels in Critically Ill Patients with AF. Stratified restricted cubic spline analysis of stress hyperglycemia ratio and mortality by SOFA severity: **a** In-hospital, **b** 30-day, **c** 365-day all-cause mortality

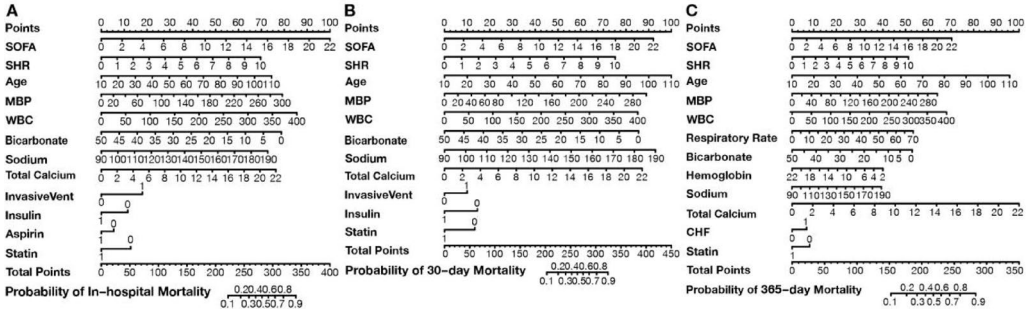


Fig. 2 Nomograms for Predicting Mortality Risks in Critically Ill AF Patients. These three nomograms predict **a** In-hospital, **b** 30-day, **c** 365-day all-cause mortality risks in critically ill patients with atrial fibrillation. Each nomogram incorporates the SHR, SOFA score, and other clinical indicators to provide a risk assessment tool

Similarly, non-diabetic patients in this high-risk group showed markedly increased mortality (HR 5.07, 95% CI 3.78–6.80). The risk patterns remained consistent across sex, acute myocardial infarction, heart failure, and acute kidney injury subgroups, with all confidence intervals excluding the null value except for a few underpowered subgroups.

HR, hazard ratio.

Model I: Unadjusted model.

Model II: Adjusted For Age; Gender; Heart Rate; Respiratory Rate; SAPSII; SIRS; PT; Sodium; Total Calcium; Potassium; Creatinine; PLT; WBC; Bicarbonate; Mean Blood Pressure; Congestive Heart Failure; Diabetes; AKI; Invasive Ventilation, Insulin; Aspirin; Statin.

SHR: Tertiles 1 (< 0.93), Tertiles 2 (0.93 ≤ SHR < 1.22), Tertiles 3 (SHR ≥ 1.22). SOFA: Low (SOFA score < 5), High (SOFA score ≥ 5).

The adjusted RCS analysis revealed a varying dose response pattern of SHR on mortality among SOFA groups and at different time intervals (Figure1). In the low SOFA group, SHR had a nonlinear correlation with in-hospital ($P_{nonlinear}$=0.001) and 30-day ($P_{nonlinear}$=0.003) mortality, marked by threshold effects at SHR = 1.0. Nonetheless, this nonlinear association was not statistically significant for 1-year mortality ($P_{nonlinear}$=0.168). In contrast, within the high SOFA group, SHR had a substantial correlation with mortality at all intervals (in-hospital, 30 days, and 1 year), characterized by a persistent J-shaped curve ($P_{nonlinear}$<0.001) and an ideal SHR of 1.2 (P < 0.001).

Mortality risk prediction based on nomogram model

Feature selection was conducted by backward stepwise regression analysis (retention significance level: α = 0.05), which determined the optimum covariate combination for each mortality outcome. Comprehensive mortality risk prediction nomograms were then developed utilizing multivariable logistic regression for three pivotal timepoints: in-hospital, 30-day, and 1-year mortality (Figure 2). The stepwise-derived models consistently maintained essential predictors at all time points, including SHR, SOFA score, age, with time-specific inclusions. The stepwise elimination approach removed non-significant variables (P ≥ 0.05) at each stage, resulting in concise

yet clinically useful models. Each retained predictor was allocated a points scale (0-100) in accordance with its regression coefficient, with SOFA consistently exhibiting the greatest contribution (maximum of 22 points). The total points system ranged from 0 to 400 points, nonlinearly correlating with mortality probability from 0.1 to 0.9 via logistic transformation. Internal validation through 1,000 bootstrap resamples demonstrated exceptional calibration across all timepoints: for in-hospital mortality, the optimism-corrected calibration curve exhibited 94–98% overlap between observed and predicted probabilities (mean absolute error [MAE] = 0.020, 90% quantile error = 0.033); for 30-day mortality, the curve sustained 90–94% concordance (MAE = 0.009, 90% quantile error = 0.012); and for 1-year mortality, calibration remained robust with 90–94% agreement (MAE = 0.021, 90% quantile error = 0.032). In all calibration plots (Figure S3), the optimism-corrected curves closely aligned with both the ideal reference line and observed outcomes, indicating low overfitting and strong predicted accuracy across the risk spectrum. The models also demonstrated robust discriminatory ability, with C-indexes of 0.807 (95% CI: 0.786–0.828) for in-hospital mortality, 0.792 (95% CI: 0.773–0.812) for 30-day mortality, and 0.722 (95% CI: 0.709–0.735) for 1-year mortality.

Model interpretation

SHAP values were employed to assess the influence of clinical variables on the predictive mortality model for critically ill atrial fibrillation patients. The SHAP summary plots (Figure 3A-C) demonstrated distinctly different patterns of influence for each feature over several time intervals: in-hospital, 30-day, and 365-day mortality. At all-time intervals, the SHR was consistently a significant predictor of mortality, with a positive SHAP indicating a higher mortality risk. In-hospital mortality was associated with protective effects from statin and aspirin use, while bicarbonate and MBP levels were identified as risk variables. The SHAP analysis of 30-day mortality indicated a similar trend. It was revealed that statin and insulin treatment conferred protection, whereas bicarbonate and sodium levels constituted risk factors. In the context of 365-day Death Prediction, statin use was correlated with reduced SHAP values, suggesting a protective influence on long-term mortality. Conversely, elevated SHAP values were noted for MBP, respiratory rate, and total calcium, which were correlated with increased risk. SHAP dependence plots elucidated nonlinear mortality associations, identifying the clinically significant SHR risk categories using a zero SHAP value as the risk-neutral benchmark: a high-risk range (SHAP > 0) emerged beyond SHR > 1.2, demonstrating exponential risk escalation when accompanied by organ dysfunction (SOFA > 5, yellow points) (Figure 3D-F).

Mediation analysis

Mediation research assessed the role of bicarbonate depletion in elucidating the mortality risk linked to the

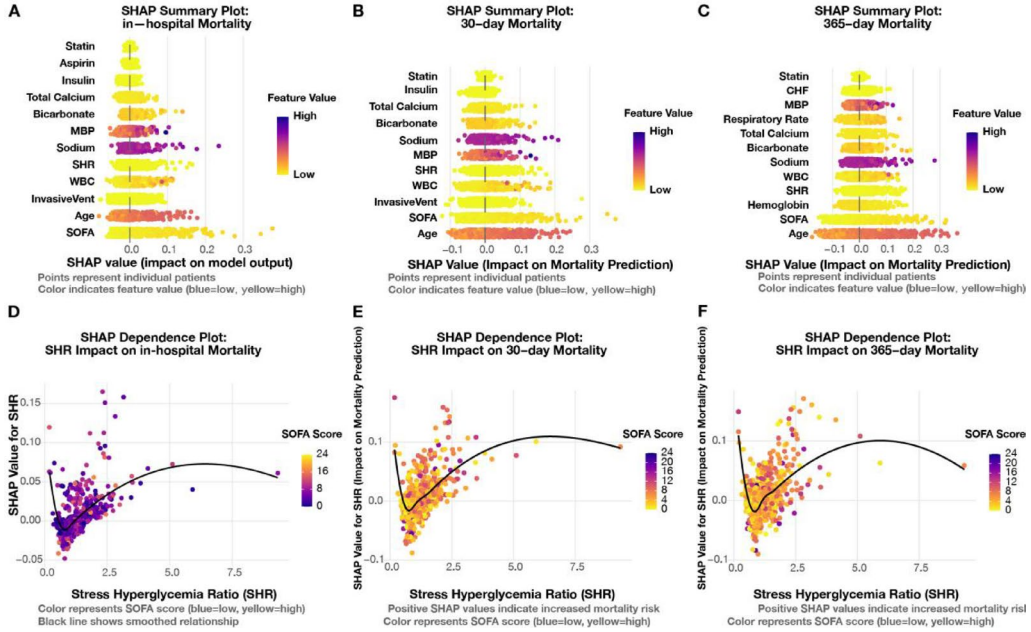


Fig. 3 SHAP Analysis of Clinical Features Influencing Mortality Risk at Different Time Intervals. SHAP summary plots (**a–c**) and SHAP dependence plots (**d–f**) are displayed, illustrating the impact of various clinical features on the prediction of in-hospital, 30-day, and 365-day mortality in critically ill patients with AF. The summary plots rank features by their SHAP values, highlighting their relative importance in mortality prediction. The dependence plots further explore the nonlinear relationship between the SHR and mortality risk across different SOFA score levels

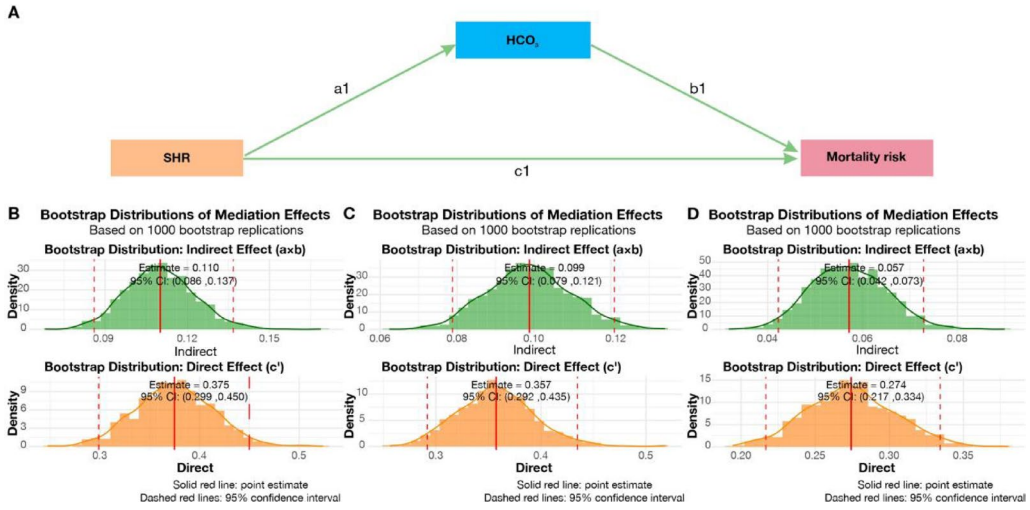


Fig. 4 Mediation analysis of the impact of SHR on mortality risk via bicarbonate. Panel A illustrates the mediation model, with panels B, C, and D showing the bootstrap distributions of the indirect (a×b) and direct effects (c') for in-hospital, 30-day, and 365-day mortality predictions, respectively

SHR. Bicarbonate accounted for 22.7% of SHR's influence on in-hospital mortality (indirect effect a × b = 0.110, 95% CI: 0.086–0.137; direct effect c' = 0.375, 95% CI: 0.299–0.450). At the 30-day follow-up, the mediation fraction diminished to 21.7% (axb = 0.099, 0.079–0.121; c' = 0.357, 0.292–0.435), but at the 1-year endpoint, merely 17.2% of the mortality impact of SHR was mediated through acid-base pathways (a × b = 0.057, 0.042–0.078; c' = 0.274, 0.217–0.334). The gradual reduction of bicarbonate mediating function corresponded with the diminishing clinical importance of metabolic acidosis in long-term critical care results (Figure 4).

Discussion

This study demonstrates that SHR significantly modulates the relationship between SOFA-defined organ failure and mortality risk in critically ill AF patients. We further identified bicarbonate-mediated pathways and developed predictive models using explainable machine learning. The interplay between SHR and SOFA scores profoundly influences mortality risk, emphasizing the necessity of their combined assessment in clinical practice. Our findings highlight the crucial interplay between metabolic stress and organ dysfunction in AF patients, but why is SHR particularly relevant in this specific population?

The implementation of SHR holds particular significance in critically ill patients with AF for several compelling reasons. First, the heightened metabolic demand and catecholamine surge inherent in AF—especially with rapid ventricular response—can exacerbate stress hyperglycemia, creating a vicious cycle that further promotes arrhythmogenesis [21, 22]. Second, the pro-inflammatory and pro-fibrotic pathways activated by stress hyperglycemia directly contribute to the atrial structural remodeling that sustains AF [23, 24]. Third, there exists a well-established pathophysiological interplay between glucose metabolism and AF, where insulin resistance and glycemic fluctuations have been shown to alter atrial electrophysiology and promote atrial ectopy [13, 25]. In this context, SHR serves as an integrated marker of this maladaptive metabolic-stress response. Unlike isolated glucose measurements, SHR accounts for underlying glycemic status, thereby more accurately reflecting the true magnitude of acute metabolic disturbance and its impact on the vulnerable atrial substrate [9, 26]. Consequently, SHR provides a superior tool for risk stratification in this specific population, identifying those AF patients in whom metabolic dysregulation may be driving both arrhythmia persistence and organ dysfunction.

SHR is associated with cardiovascular risk factors, including IR, and may serve as an important clinical prognostic indicator in cardiovascular diseases [26]. SHR is commonly observed in patients with critical illnesses and is closely linked to inflammatory reactions and significant disturbances in glucose metabolism [23, 27, 28]. Clinical studies indicate that SHR is independently linked to numerous poor clinical outcomes, including thrombosis load and coronary artery disease severity [29, 30]. In individuals suffering from AMI, a heightened SHR demonstrated a significant association with long-term all-cause mortality within cohorts from both the United States and China [31]. Conversely, among patients experiencing acute and decompensated heart failure, SHR revealed a U-shaped relationship with both mortality and rehospitalization rates [7].

Furthermore, Cheng et al. proposed a U-shaped correlation between the levels of the SHR index and mortality rates observed at 30, 90, 180, and 365 days [14]. Nonetheless, the impact of SHR on mortality risk or the severity

of organ failure has yet to be investigated. This study elucidates that SHR dynamically interacts with SOFA-defined organ failure, accordingly, altering death trajectories. Upon controlling for potential confounders, there discerned a threshold-dependent J-shaped correlation between SHR and in-hospital or 30-day mortality in critically sick AF patients with SOFA > 5, with an inflection point at SHR = 1.2. In contrast, this connection exhibited a J-shaped pattern during the acute period in low-SOFA patients (SOFA ≤ 5), but it became statistically insignificant for 365-day mortality. The findings of this study demonstrate that the prognostic value of SHR is dependent upon concurrent organ dysfunction, presenting a stratified risk assessment framework that was previously unavailable.

The precise mechanisms that contribute to the nonlinear relationship are not yet fully understood. Hyperglycemia triggered by stress manifests during stress or significant illness [32]. An elevated SHR suggests that the body's stress response remains unresolved, indicating sustained autonomic nervous system dysfunction. This condition, characterized by overactivity of the sympathetic nervous system and reduced parasympathetic nervous system activity, may facilitate AF development [21, 22]. Moreover, acute hyperglycemia excursions induce oxidative stress by promoting the overproduction of reactive oxygen species [25]. Thirdly, a high SHR may interfere with fibrinolysis, which breaks down thrombi, characterized by elevated levels of plasminogen activator inhibitor, indicating a direct correlation between hyperglycemia and hypofibrinolytic conditions [33, 34]. Simultaneously, when the SHR is elevated, endothelial dysfunction occurs because of a mismatch between vasodilatory and vasoconstrictive factors (such as endothelin and angiotensin II) [35]. The fibrinolytic and endothelial abnormalities jointly enhance platelet adhesion and aggregation [36]. In addition, sustained hyperglycemia significantly adds to non-cardiovascular mortality by hindering tissue healing, elevating infection risk, and extending hospitalization duration.

It is still unclear what precisely explains the additive effect. The J-shaped mortality curve associated with SHR (inflection point: 1.0 or 1.2) reflects a fundamental metabolic transition: below this threshold, moderate hyperglycemia preserves hemodynamic stability through catecholamine-mediated vasoconstriction [37], while exceeding it leads to mitochondrial oxidative stress and endothelial damage, culminating in metabolic acidosis—the primary pathophysiological mechanism driving SHR-related mortality [38, 39]. In critically ill AF patients, an elevated SHR intensifies multi-organ failure via specific cardiometabolic mechanisms. SHR-induced metabolic stress diminishes myocardial bioenergetics via decreasing cardiomyocyte adenosine triphosphate (ATP) production, undermining heart function within the SOFA framework. Advanced glycol end-products (e.g. AGEs) facilitate pro-fibrotic atrial remodeling [24], whereas adrenergic induced left atrial dilation enhances mechanical stiffness [25]. These mechanisms cumulatively exacerbate the effects of pre-existing organ failure, especially via heightened systemic inflammation and acidosis—by impairing inter-organ communication. This interaction between metabolic dysregulation and SOFA-defined organ dysfunction increases mortality risk, establishing SHR as a significant regulator of organ failure pathogenesis in critical care environments.

Importantly, this study newly evaluated the mediation analysis quantifying this axis: bicarbonate levels accounted for 22.7% of in-hospital deaths and 17.2% of long-time mortality, establishing acidosis as the mechanistic linchpin between hyperglycemic stress and organ failure. The hierarchical mediation effect may be mediated by three different pathways: firstly, there is a bioenergetic collapse caused by acidosis, which inhibits cytochrome C oxidase, resulting in a reduction in cardiac ATP synthesis, resulting in a worsening of SOFA-cardiovascular scores; secondly, renal compensation occurs when sustained acidosis triggers SLC26A4-mediated bicarbonate reabsorption [40], resulting in a decline over time in mediation proportions; additionally, a combination of acidotic environments and coagulopathy exacerbation leads to an increase in platelet hyper aggregation and PAI-1 elevation, which synergizes with the SOFA-coagulation sub scores to drive the mortality interaction [41]. In summary, the effects of acidotic environments are amplified by glycocalyx shedding and platelet hyper aggregation, amplifying the synergistic effects of SHR and SOFA on mortality. While bicarbonate demonstrated significant mediation, other mechanisms likely contribute to the SHR-mortality relationship. Inflammatory markers and indicators of endothelial dysfunction—not consistently captured in our dataset—may represent important additional pathways [22, 35, 38]. This suggests that bicarbonate represents one component of a more complex pathway. Future studies with comprehensive biomarker profiling are warranted to fully elucidate the underlying mechanisms.

Conventional ICU scoring systems such as SAPS II and SOFA, although useful for general risk evaluation [33, 42], demonstrate significant shortcomings in populations with atrial fibrillation. This is attributable to their formulation in non-cardiac groups and their failure to account for AF-specific pathophysiology, especially the intricate relationship between metabolic stress and organ dysfunction [43]. To overcome these limitations, this study created a metabolically graded intervention paradigm based on the relationship between SHR thresholds and SOFA scores, where SHR consistently ranked among the top

5 contributors to mortality prediction across all timeframes, outperforming conventional markers like blood pressure and electrolytes. Crucially, this synergy explains why existing ICU scoring systems (e.g., SOFA, SAPS II) fail in AF cohorts: they overlook the bidirectional pathophysiology where hyperglycemia-induced mitochondrial bioenergetic failure and coagulopathy accelerate organ damage, while progressive organ dysfunction further exacerbates metabolic acidosis. For patients in the Protected tier (SHR < 1.2 + SOFA < 5; SHAP < 0), sustaining the existing glycemic control is enough due to the negligible risk of mortality associated with acidosis. Individuals classified in the Alert tier (SHR 1.2–2.5; SHAP 0-0.2) necessitate enhanced bicarbonate surveillance and insulin adjustment to mitigate emerging acidotic damage—thereby averting the 20% mediational mortality identified ins our study. Patients advancing to the critical tier (SHR > 2.5 + SOFA > 8; SHAP > 0.2) necessitate immediate metabolic correction, disrupting the self-reinforcing cycle of mitochondrial bioenergetic failure and coagulopathy that exacerbates mortality risk.

Strengths and limitations

The present study introduces an innovative framework for managing AF in critical care settings, revealing that the SHR has an active effect on mortality associated with organ failure caused by acidotic injury, a mechanism that had not been considered by traditional ICU scoring systems. By integrating interpretable machine learning, mediation analysis, and time-stratified survival modeling, this research addresses a critical weakness of current prognostic methods such as SOFA and SAPS II: their failure to account for atrial fibrillation-specific metabolic-organ interactions. A clinically actionable, metabolically graded intervention framework has been established which surpasses correlation-based risk prediction. Based on SHAP-derived thresholds and validated through multi-modal analysis, this framework categorizes patients into three distinct physiological groups—Protected, Alert, and Critical—that require a variety of therapeutic approaches from watchful waiting to continuous renal replacement therapy. This study redefines atrial fibrillation care in the ICU by emphasizing acid-base homeostasis in tandem with anticoagulation-transforming clinical practice from reactive organ support to proactive metabolic intervention.

There are, however, some limitations to this study. First, the retrospective design using the MIMIC-IV database precludes causal inference, and temporal changes in clinical practice over the long study period may affect the generalizability of our findings to contemporary settings. Furthermore, the single center data may affect generalizability to other healthcare settings with different patient populations, management protocols, or resource availability. Second, the absence of specific biomarkers—such as inflammatory markers, echocardiographic parameters, and direct measures of mitochondrial dysfunction (e.g., ATP) or coagulopathy (e.g., PAI-1)—limits its mechanistic interpretation. Third, and critically, the database does not allow for detailed classification of atrial fibrillation, including its etiology (primary cause of admission versus secondary complication), type (paroxysmal or persistent), or chronicity (new-onset versus pre-existing). Analysis of treatment strategies such as rate versus rhythm control was also not feasible. Despite these constraints, our model relies on readily available clinical variables and remains broadly applicable to the heterogeneous population of critically ill AF patients. Further validation in prospectively defined subgroups with better phenotyping is warranted.

Future research could leverage advanced computational methods to enhance dynamic risk prediction. For instance, later temporal attention mechanisms could capture the evolving relationship between SHR trajectories and organ dysfunction over the ICU stay, potentially identifying critical windows for intervention. Similarly, singular pooling techniques could optimize feature representation from high-frequency physiological data, improving model efficiency and interpretability for real-time clinical decision support in critical care settings.

Conclusion

This study illustrates that the SHR alters the relationship between the severity of organ failure and mortality in critically sick patients with atrial fibrillation, with acidosis-related mechanisms being partially mediated by serum bicarbonate concentrations. The use of SHR with SOFA scores enhances mortality risk categorization beyond traditional ICU prognostic models. Regular monitoring of SHR and bicarbonate is advised for this demographic, with SHR thresholds integrated into SOFA assessments to inform metabolic-targeted treatments. These findings advocate for the inclusion of acute hyperglycemia treatment in organ protection strategies for critical care atrial fibrillation.

Abbreviations

SOFA Sequential Organ Failure Assessment
SAPS II Simplified Acute Physiology Score II
SIRS Systemic Inflammatory Response Syndrome Score
OASIS Outcome and Severity Index Score

Supplementary Information

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Supplementary Material 1

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Author contributions
HJ, XF, BZ, and JL played pivotal roles in the design and conceptual framework of the research study. The initial draft of the manuscript was authored by HJ, who received subsequent input from XF, BZ, and JL. HJ also conducted the statistical analyses for the study. All authors were involved in the interpretation of the data, meticulously reviewed the manuscript, and consented to the publication of the final version.

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Data availability
Any inquiries pertaining to the sharing of datasets or methodologies should be directed to the corresponding author. The publicly accessible MIMIC datasets can be found at https://mimic.mit.edu/.

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Al-Khatib SM. Atrial fibrillation. Ann Intern Med. 2023;176(7):Itc97–112.
2. Staerk L, Wang B, Preis SR, Larson MG, Lubitz SA, Ellinor PT, McManus DD, Ko D, Weng LC, Lunetta KL, et al. Lifetime risk of atrial fibrillation according to optimal, borderline, or elevated levels of risk factors: cohort study based on longitudinal data from the Framingham heart study. BMJ. 2018;361:k1453.
3. Wetterslev M, Haase N, Hassager C, Belley-Cote EP, McIntyre WF, An Y, Shen J, Cavalcanti AB, Zampieri FG, Guimaraes HP, et al. New-onset atrial fibrillation in adult critically ill patients: a scoping review. Intensive Care Med. 2019;45(7):928–38.
4. Wetterslev M, Hylander Møller M, Granholm A, Hassager C, Haase N, Lange T, Myatra SN, Hästbacka J, Arabi YM, Shen J, et al. Atrial fibrillation (AFIB) in the ICU: Incidence, risk Factors, and outcomes: the international AFIB-ICU cohort study. Crit Care Med. 2023;51(9):1124–37.
5. Moss TJ, Calland JF, Enfield KB, Gomez-Manjarres DC, Ruminski C, DiMarco JP, Lake DE, Moorman JR. New-Onset atrial fibrillation in the critically ill. Crit Care Med. 2017;45(5):790–7.
6. Raghavan S, Vassy JL, Ho YL, Song RJ, Gagnon DR, Cho K, Wilson PWF, Phillips LS. Diabetes Mellitus-Related All-Cause and cardiovascular mortality in a National cohort of adults. J Am Heart Assoc. 2019;8(4):e011295.
7. Zhou Q, Yang J, Wang W, Shao C, Hua X, Tang YD. The impact of the stress hyperglycemia ratio on mortality and rehospitalization rate in patients with acute decompensated heart failure and diabetes. Cardiovasc Diabetol. 2023;22(1):189.
8. Yang J, Zheng Y, Li C, Gao J, Meng X, Zhang K, Wang W, Shao C, Tang YD. The impact of the stress hyperglycemia ratio on Short-term and Long-term poor prognosis in patients with acute coronary syndrome: insight from a large cohort study in Asia. Diabetes Care. 2022;45(4):947–56.
9. Roberts GW, Quinn SJ, Valentine N, Alhawassi T, O'Dea H, Stranks SN, Burt MG, Doogue MP. Relative hyperglycemia, a marker of critical illness: introducing the stress hyperglycemia ratio. J Clin Endocrinol Metab. 2015;100(12):4490–7.
10. Song Y, Cui K, Yang M, Song C, Yin D, Dong Q, Gao Y, Dou K. High triglyceride-glucose index and stress hyperglycemia ratio as predictors of adverse cardiac events in patients with coronary chronic total occlusion: a large-scale prospective cohort study. Cardiovasc Diabetol. 2023;22(1):180.
11. Cui K, Fu R, Yang J, Xu H, Yin D, Song W, Wang H, Zhu C, Feng L, Wang Z, et al. The impact of fasting stress hyperglycemia ratio, fasting plasma glucose and hemoglobin A1c on in-hospital mortality in patients with and without diabetes: findings from the China acute myocardial infarction registry. Cardiovasc Diabetol. 2023;22(1):165.
12. Luo J, Li Z, Qin X, Zhang X, Liu X, Zhang W, Xu W, Liu B, Wei Y. Association of stress hyperglycemia ratio with in-hospital new-onset atrial fibrillation and long-term outcomes in patients with acute myocardial infarction. Diabetes Metab Res Rev. 2024;40(2):e3726.
13. Papazoglou AS, Kartas A, Moysidis DV, Tsagkaris C, Papadakos SP, Bekiaridou A, Samaras A, Karagiannidis E, Papadakis M, Giannakoulas G. Glycemic control and atrial fibrillation: an intricate relationship, yet under investigation. Cardiovasc Diabetol. 2022;21(1):39.
14. Cheng S, Shen H, Han Y, Han S, Lu Y. Association between stress hyperglycemia ratio index and all-cause mortality in critically ill patients with atrial fibrillation: a retrospective study using the MIMIC-IV database. Cardiovasc Diabetol. 2024;23(1):363.
15. Johnson AEW, Bulgarelli L, Shen L, Gayles A, Shammout A, Horng S, Pollard TJ, Hao S, Moody B, Gow B, et al. MIMIC-IV, a freely accessible electronic health record dataset. Sci Data. 2023;10(1):1.
16. Montez-Rath ME, Winkelmayer WC, Desai M. Addressing missing data in clinical studies of kidney diseases. Clin J Am Soc Nephrol. 2014;9(7):1328–35.
17. Stekhoven DJ, Bühlmann P. MissForest–non-parametric missing value imputation for mixed-type data. Bioinformatics. 2012;28(1):112–8.
18. Kohnke J, Pattberg K, Nensa F, Kuhlmann H, Brenner T, Schmidt K, Hosch R, Espeter F. A proof of concept for microcirculation monitoring using machine learning based hyperspectral imaging in critically ill patients: a monocentric observational study. Crit Care. 2024;28(1):230.
19. Cook NR. Statistical evaluation of prognostic versus diagnostic models: beyond the ROC curve. Clin Chem. 2008;54(1):17–23.
20. Lundberg S, Lee SI. A Unified Approach to Interpreting Model Predictions. 2017.
21. Chen PS, Chen LS, Fishbein MC, Lin SF, Nattel S. Role of the autonomic nervous system in atrial fibrillation: pathophysiology and therapy. Circ Res. 2014;114(9):1500–15.
22. Jernås M, Olsson B, Sjöholm K, Sjögren A, Rudemo M, Nellgård B, Carlsson LM, Sjöström CD. Changes in adipose tissue gene expression and plasma levels of adipokines and acute-phase proteins in patients with critical illness. Metabolism. 2009;58(1):102–8.
23. Qin X, Zhang Y, Zheng Q. Metabolic inflexibility as a pathogenic basis for atrial fibrillation. Int J Mol Sci 2022, 23(15).
24. Zelniker TA, Jarolim P, Scirica BM, Braunwald E, Park JG, Das S, Sabatine MS, Morrow DA. Biomarker of collagen turnover (C-Terminal Telopeptide) and prognosis in patients with Non- ST -Elevation acute coronary syndromes. J Am Heart Assoc. 2019;8(9):e011444.
25. Wang A, Green JB, Halperin JL, Piccini JP. Atrial fibrillation and diabetes mellitus: JACC review topic of the week. J Am Coll Cardiol. 2019;74(8):1107–15.
26. Sia CH, Chan MH, Zheng H, Ko J, Ho AF, Chong J, Foo D, Foo LL, Lim PZ, Liew BW, et al. Optimal glucose, HbA1c, glucose-HbA1c ratio and stress-hyperglycaemia ratio cut-off values for predicting 1-year mortality in diabetic and non-diabetic acute myocardial infarction patients. Cardiovasc Diabetol. 2021;20(1):211.
27. Cui K, Fu R, Yang J, Xu H, Yin D, Song W, Wang H, Zhu C, Feng L, Wang Z, et al. Stress hyperglycemia ratio and long-term mortality after acute myocardial infarction in patients with and without diabetes: A prospective, nationwide, and multicentre registry. Diabetes Metab Res Rev. 2022;38(7):e3562.
28. Hill MA, Yang Y, Zhang L, Sun Z, Jia G, Parrish AR, Sowers JR. Insulin resistance, cardiovascular stiffening and cardiovascular disease. Metabolism. 2021;119:154766.
29. Zhang Y, Song H, Bai J, Xiu J, Wu G, Zhang L, Wu Y, Qu Y. Association between the stress hyperglycemia ratio and severity of coronary artery disease under different glucose metabolic States. Cardiovasc Diabetol. 2023;22(1):29.
30. Algül E, Özbeyaz NB, Şahan HF, Aydınyılmaz F, Sunman H, Tulmaç M. Stress hyperglycemia ratio is associated with high thrombus burden in patients with acute coronary syndrome. Angiology. 2024;75(7):645–50.
31. Liu J, Zhou Y, Huang H, Liu R, Kang Y, Zhu T, Wu J, Gao Y, Li Y, Wang C, et al. Impact of stress hyperglycemia ratio on mortality in patients with critical acute myocardial infarction: insight from American MIMIC-IV and the Chinese CIN-II study. Cardiovasc Diabetol. 2023;22(1):281.

32. Shamoon H, Hendler R, Sherwin RS. Synergistic interactions among antiinsulin hormones in the pathogenesis of stress hyperglycemia in humans. J Clin Endocrinol Metab. 1981;52(6):1235–41.
33. Lemkes BA, Hermanides J, Devries JH, Holleman F, Meijers JC, Hoekstra JB. Hyperglycemia: a prothrombotic factor? J Thromb Haemost. 2010;8(8):1663–9.
34. Cefalu WT, Schneider DJ, Carlson HE, Migdal P, Gan Lim L, Izon MP, Kapoor A, Bell-Farrow A, Terry JG, Sobel BE. Effect of combination glipizide GITS/metformin on fibrinolytic and metabolic parameters in poorly controlled type 2 diabetic subjects. Diabetes Care. 2002;25(12):2123–8.
35. Ferroni P, Basili S, Falco A, Davì G. Platelet activation in type 2 diabetes mellitus. J Thromb Haemost. 2004;2(8):1282–91.
36. Vinik AI, Erbas T, Park TS, Nolan R, Pittenger GL. Platelet dysfunction in type 2 diabetes. Diabetes Care. 2001;24(8):1476–85.
37. Song G, Liu X, Lu Z, Guan J, Chen X, Li Y, Liu G, Wang G, Ma F. Relationship between stress hyperglycaemic ratio (SHR) and critical illness: a systematic review. Cardiovasc Diabetol. 2025;24(1):188.
38. Diaz-Juarez J, Suarez JA, Dillmann WH, Suarez J. Mitochondrial calcium handling and heart disease in diabetes mellitus. Biochim Biophys Acta Mol Basis Dis. 2021;1867(1):165984.
39. El Hadi H, Vettor R, Rossato M. Cardiomyocyte mitochondrial dysfunction in diabetes and its contribution in cardiac arrhythmogenesis. Mitochondrion. 2019;46:6–14.
40. Alsufayan TA, Myers EJ, Quade BN, Brady CT, Marshall A, Haque N, Duffey ME, Parker MD. Revisiting the role of Ser982 phosphorylation in stoichiometry shift of the electrogenic Na(+)/qHCO(3)(-) cotransporter NBCe1. Int J Mol Sci 2021, 22(23).
41. Monnier L, Mas E, Ginet C, Michel F, Villon L, Cristol JP, Colette C. Activation of oxidative stress by acute glucose fluctuations compared with sustained chronic hyperglycemia in patients with type 2 diabetes. JAMA. 2006;295(14):1681–7.
42. Lambden S, Laterre PF, Levy MM, Francois B. The SOFA score-development, utility and challenges of accurate assessment in clinical trials. Crit Care. 2019;23(1):374.
43. Chen Y, Huang B, Calvert P, Liu Y, Gue Y, Gupta D, McDowell G, Azariah JL, Namboodiri N, Unni G, et al. Phenotypes of South Asian patients with atrial fibrillation and holistic integrated care management: cluster analysis of data from KERALA-AF registry. Lancet Reg Health Southeast Asia. 2024;31:100507.

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